

METALLIC PHASES PRECIPITATED IN UO_2 FUEL

II. Insoluble residue in simulated fuel

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The formation and the chemical properties of the insoluble residue in simulated spent fuels were studied. Simulated fuels corresponding to burnups of 5 to 30% FIMA were prepared by adding non-radioactive FP elements to uranium in nitrate solution, evaporating the mixed solution to dryness and heating the pellet at 1273 to 2273 K under various oxygen potentials. The insoluble residue was obtained at the dissolution of the simulated fuel with 3M HNO_3 . The chemical composition of the residue was determined by chemical analysis and the phases in the residue were identified by X-ray diffraction. The insoluble residue was composed mainly of Mo and Ru with minor amounts of Pd and Rh. At some conditions Zr and Sn were also included. The amount of residue was less than 1 wt% up to a 10% FIMA burnup, but increased markedly at a higher burnup. The amount of residue decreased at a temperature higher than 1873 K. This was attributed to the fact that the metallic α -phase formed at high temperature was apt to be easily dissolved in HNO_3 . The amount of metallic residue and Mo/Ru ratio decreased with the increase in oxygen potential. The oxygen potential for Mo– MoO_2 equilibrium may be related to this result. Formation of the Zr-molybdate hydrated oxide compound, $\text{ZrMo}_2\text{O}_7(\text{OH})_2(\text{H}_2\text{O})_2$, was observed only in the residue at high burnups and high oxygen potentials. This phase was possibly formed during the dissolution process.

1. Introduction

It is well known that insoluble materials remain during dissolution of spent nuclear fuel in dilute HNO_3 solution for reprocessing and that they increase in amount as the nuclear fuel is burned up for a longer period [1,2]. These consist of metallic phases formed in the fuel during irradiation, reprecipitates from the spent fuel solution, crud (radioactive corrosion products) adhering on the surface of fuel pins, fine chips of cladding material formed during chopping of the fuel pins, etc [2]. As the increased amount of the insoluble materials may cause some problems during reprocessing at the clarification and extraction process [3,4] such as degradation of the solvent due to their high radioactivity, deterioration of decontamination factor and clogging of the piping system of the equipment, it is thus important to know how the amount of residue and its chemical composition are related to such parameters as burnup, oxygen potential and temperature during irradiation.

In our previous paper, oxide and metallic phases have been prepared by the reaction between UO_2 and simulated fission products corresponding to 5 to 30% FIMA at temperatures of 1273–2273 K under various

oxygen potentials, and the conditions for their formation have been examined [5]. By X-ray diffraction, a hexagonal structure phase (ϵ -phase) and a cubic structure phase (α -phase) were identified in the simulated fuels before dissolution with HNO_3 . After dissolution, the α -phase was not observed in the residue, whereas the ϵ -phase still remained. However, a tetragonal structure phase (σ -phase) was identified only in the insoluble residue derived from the fuel under an atmosphere of approximately -300 KJ/mol O_2 . It has been reported by many researchers that the hexagonal ϵ -phase is a Ru-based alloy, and that its main components are Ru, Mo, Tc and Pd [6,7]. It was found in our previous paper that lattice parameters of the hexagonal ϵ -phase at 1673 K sharply decreased at oxygen potentials larger than -300 kJ/mol O_2 . This may be explained from the fact that the Mo– MoO_2 equilibrium has an oxygen potential of about -300 kJ/mol O_2 at 1673 K. With the increase of the oxygen potential above -300 kJ/mol O_2 , the Mo/ MoO_2 ratio in the fuel decreased. Therefore, both Mo content in the hexagonal phase and the lattice parameters of the ϵ -phase decreased. Lattice parameters obtained for the tetragonal σ -phase were in good agreement with those of Mo_5Ru_3 [5]. This phase was identi-

fied in the samples treated either at 1673 K with oxygen potential near to Mo–MoO₂ equilibrium or at 2073 K with low oxygen potential. The dependence of the lattice parameters of the tetragonal σ -phase on the temperature and oxygen potential at sample preparation were relatively small.

In the present paper, UO₂ fuels simulated for use in light water reactors with burnups of 5 to 30% FIMA were dissolved in 3M HNO₃, and the amount of insoluble residue and its elemental composition determined by chemical analysis were shown comparing the metallic phases identified by X-ray diffraction.

2. Experimental

2.1. Preparation of pellets

Compositions of simulated fuels corresponding to burnups of 5 to 30% FIMA are given in the previous paper [5].

Mixed solutions containing U and FP elements were prepared and dried on a hot plate and the residue obtained was calcined at 1173 K in a stream of 4% H₂ + 96% He. The calcine was pressed into pellets 7 mm in diameter and 1.0 g in weight. They were then heat-treated for 4 h at 1273 to 2273 K in a stream of one of the following gases: He, 50% CO₂ + 50% CO, 10% CO₂ + 90% CO or 4% H₂ + 96% He.

2.2. Dissolution of pellets and filtration of the insoluble residue

Experimental procedures are shown in fig. 1. The pellet was pulverised in an agate mortar and dissolved in 3M HNO₃ for 2 h at 368 K. The resultant solution was filtered off using a 0.1 μ m Nuclepore filter with suction. The insoluble residue thus obtained was air-dried and weighed.

2.3. Dissolution and analysis of the insoluble residue

The insoluble residue was dissolved in an acid mixture of HCl and HNO₃ (9:1) at 453 K for 20 h using a PFTE (Polytetrafluoroethylene) vial surrounded by an outer PFTE and a stainless steel vessel, as seen in fig. 2. To balance the pressure generated in the inner PFTE vial during dissolution, a small volume of H₂O was added into the outer PFTE vessel.

The dissolved solution was diluted to a definite volume and analyzed using inductively coupled plasma–atomic emission spectroscopy (ICP–AES) and

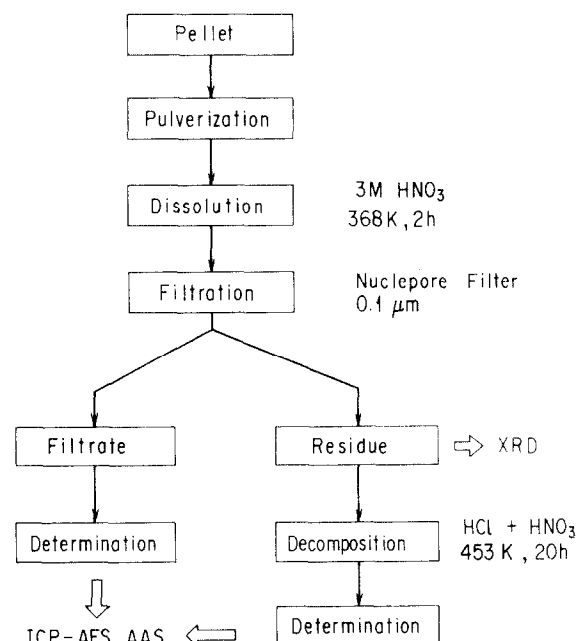


Fig. 1. Experimental procedures for the analysis of the insoluble residue.

atomic absorption spectrophotometry (AAS). X-ray diffraction (XRD) was also used to identify the phases in insoluble residue as described in our previous paper [5].

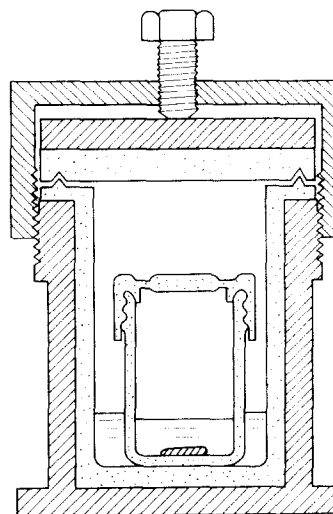


Fig. 2. Decomposition vessel for the insoluble residue made of PTFE and stainless steel.

3. Results

3.1. Effect of treatment temperature on the amount of insoluble residue and its chemical composition

The effect of temperature during heat treatment of the simulated fuel on the amount of residue derived from the dissolution of simulated fuel corresponding to 5% FIMA are shown in fig. 3 at a high (He) and a low ($4\% \text{H}_2 + 96\% \text{He}$) oxygen potential. The effect of temperature on the residue composition at the same oxygen potentials are shown in fig. 4. The chemical composition of the residue is shown in mole fraction versus treated temperature.

At a low oxygen potential controlled by $4\% \text{H}_2 + 96\% \text{He}$ (fig. 3, curve 1), the amount of insoluble residue is about 0.25 to 0.30 wt% at a temperature less than 1873 K. At a higher temperature, this begins to decrease and reaches approximately 0.1 wt% at 2273 K. In accordance with this relation, the mole fraction of Pd, as shown in fig. 4(a), begins to decrease at temperatures higher than 1873 K. In this connection, the other important tendency in composition is that the Mo/Ru ratio becomes larger at temperatures higher than 1873 K. The Mo/Ru ratio in fig. 4(a) is about 1/1 between 1273 and 1873 K and about 3/2 between 2073 and 2273 K.

On the other hand, at a high oxygen potential controlled by He, the amount of insoluble residue does not change appreciably between 1273 and 2273 K (fig. 3, curve 2). As for the elemental composition, the change of Pd mole fraction is also not clear. But, as seen at the low oxygen potential, the tendency of the Mo/Ru ratio to become larger at higher temperatures is also observed.

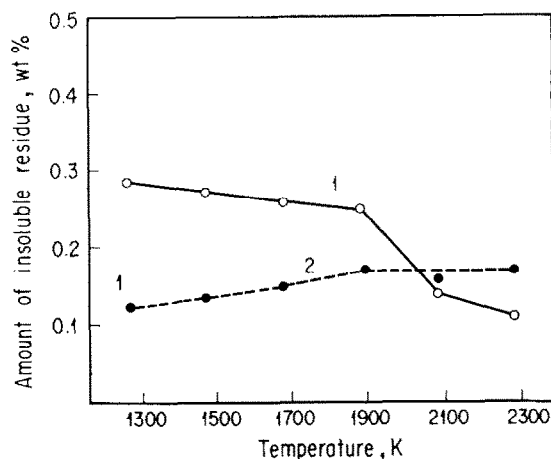
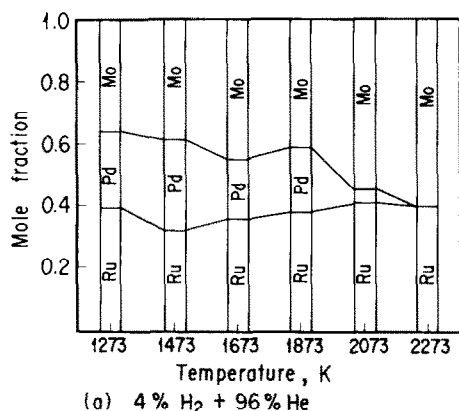


Fig. 3. Effect of heat-treatment temperature on the amount of the insoluble residue for burnup of 5% FIMA. (1): $4\% \text{H}_2 + 96\% \text{He}$; (2): He.

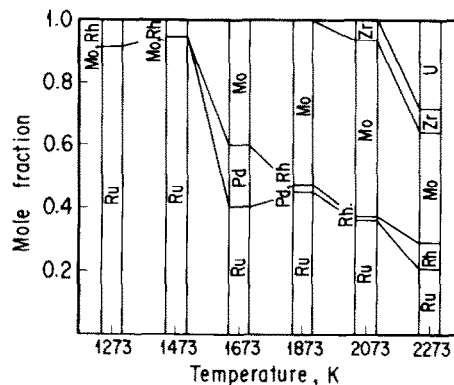
3.2. Effects of oxygen potential and burnup on the amount of insoluble residues and its composition

Changes in the amount of residue and its elemental composition with oxygen potentials are investigated for the simulated fuels of 5 to 30% FIMA heated at 1673 and 2073 K, and the results for these changes are shown in fig. 5. The effects of oxygen potential on the composition of insoluble residue are shown in fig. 6 at 5% FIMA and in fig. 7 at 10% FIMA, respectively.

The amount of insoluble residue in fig. 5 shows a maximum at the intermediate oxygen potential for both temperatures. With an increase in burnup, not only this



(a) $4\% \text{H}_2 + 96\% \text{He}$



(b) He

Fig. 4. Effect of heat-treatment temperature on the elemental composition of the insoluble residue under the atmosphere of (a) $4\% \text{H}_2 + 96\% \text{He}$ and (b) He for burnup of 5% FIMA.

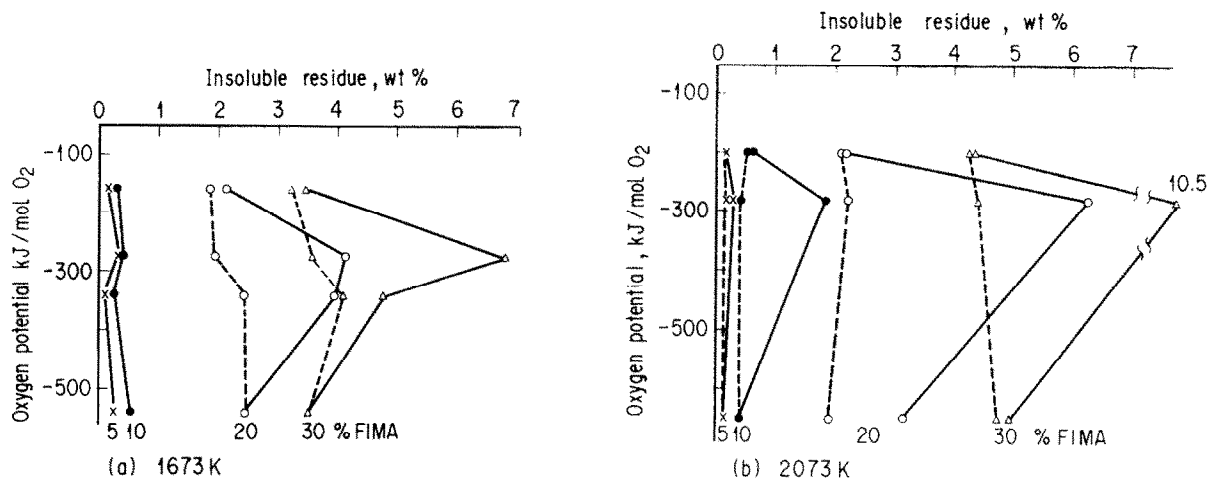


Fig. 5. Effect of oxygen potential and burnup on the amount of the insoluble residue at (a) 1673 K and (b) 2073 K. Dotted lines are corrected for the amount of $\text{ZrMo}_2\text{O}_7(\text{OH})_2(\text{H}_2\text{O})_2$.

tendency becomes clear, but also the amount of insoluble residue increases remarkably. The amount of insoluble residue is less than 1.0 wt% up to 10% FIMA, but increases 2 to 6 wt% at 20% FIMA and 3 to 10 wt% at 30% FIMA. Dotted lines are corrected for zirconium-molybdate hydrated compound, $\text{ZrMo}_2\text{O}_7(\text{OH})_3(\text{H}_2\text{O})_3$, and will be discussed later.

Only three elements (Ru, Pd and Mo) are determined in the insoluble residue obtained from the simulated fuel corresponding to burnup of 5% FIMA with heat treatment at 1673 K (fig. 6(a)). For the residue corresponding to burnup of 10% FIMA at 1673 K (fig. 7(a)),

Rh, Sn and Zr are also determined. At high heat treatment temperatures (fig. 6(b) and fig. 7(b)), Ru, Mo and Zr are the main components in the residue with minor amounts of Pd, Rh, Sn and U.

The clearest change in composition with oxygen potential takes place in the Mo/Ru ratio. This decreases with increasing oxygen potential except for the 10% FIMA fuel treated at 2073 K. As seen in fig. 7(b), a considerable amount of Zr is found in the residue obtained at higher oxygen potentials. In these residues a Zr-molybdate hydrated oxide compound, $\text{ZrMo}_2\text{O}_7(\text{OH})_2(\text{H}_2\text{O})_2$ [7], was found to exist by XRD.

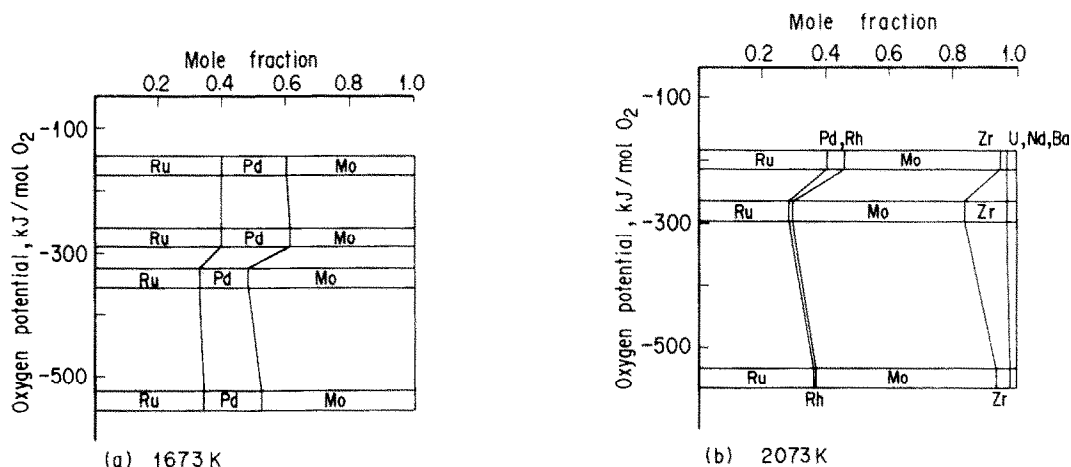


Fig. 6. Effect of oxygen potential on the elemental composition of the insoluble residue at (a) 1673 K and (b) 2073 K for burnup of 5% FIMA.

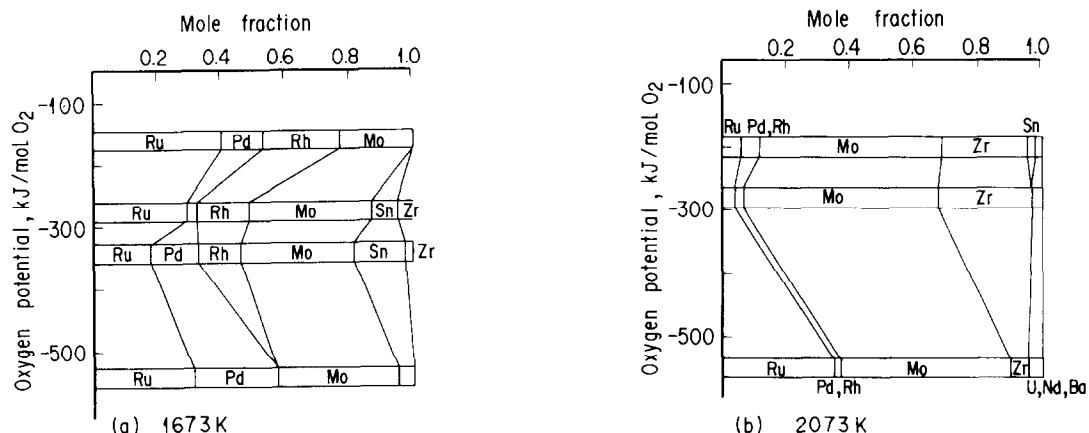


Fig. 7. Effect of oxygen potential on the elemental composition of the insoluble residue at (a) 1673 K and (b) 2073 K for burnup of 10% FIMA

If the amount of Mo combined with Zr is subtracted from the mole fraction of Mo, this leaves only 7 mol% of metallic Mo for the oxygen potential controlled by He (fig. 7(b), the highest oxygen potential). Therefore, the corrected Mo/Ru ratio also decreases with the increase in oxygen potential.

As already described concerning the effect of the temperature, Pd is found in the residue in reasonably large amounts at 1673 K, but only in scarce amounts at 2073 K both for 5 and 10% FIMA (figs. 6 and 7). A similar phenomenon is found for Sn at 10% FIMA in fig. 7.

Elemental composition for 20 and 30% FIMA was also been determined. The amount of residue is greatly

increased, but very similar results to the above-mentioned were obtained for the composition.

4. Discussion

In fig. 8, the metallic elements determined by the chemical analyses and the phases identified by XRD in the residues are shown as a function of the oxygen potential and burnup when the simulated fuel was treated at 1673 K. The circles in the figure indicate the conditions, i.e. burnup and oxygen potential of sample preparation.

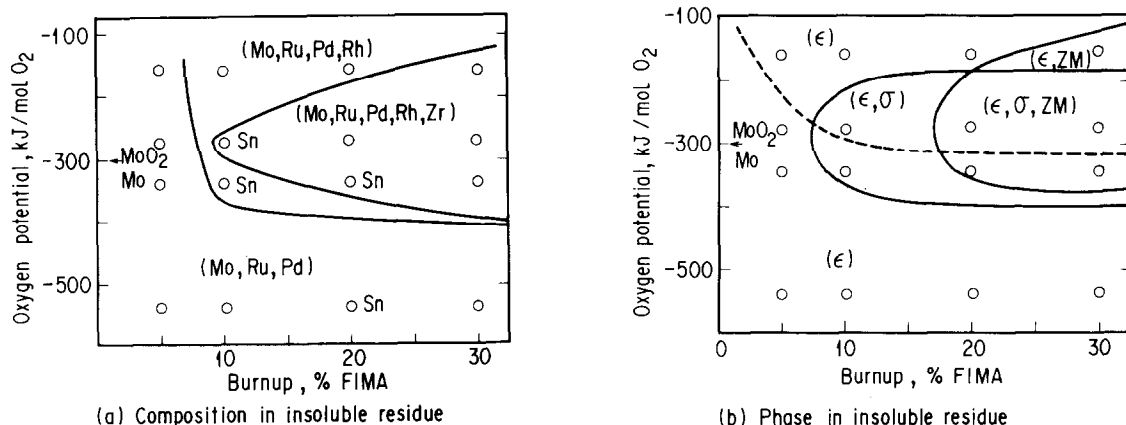


Fig. 8. The elemental composition determined by chemical analyses (a) and the phases identified by XRD (b) in the insoluble residue under various oxygen potentials and burnups at 1673 K. Above the dotted lines the α -phase is observed before dissolution with HNO_3 .

Mo, Ru, and Pd are determined at all areas, whereas Rh, Sn and Zr are determined in certain restricted areas expressed in fig. 8(a) as (Mo, Ru, Pd), (Mo, Ru, Pd, Rh) or (Mo, Ru, Pd, Rh, Zr). In addition, the points where Sn is determined are also included in fig. 8(a).

The phases identified in the residue are the ϵ -phase (hexagonal, Mo–Ru–Pd), the σ -phase (tetragonal, Mo_5Ru_3) and the hydrated oxide phase ($ZrMo_2O_7(OH)_2(H_2O)_2$, denoted as ZM) (fig. 8(b)). The areas where these phases are found are expressed as (ϵ), (ϵ , σ , ZM) etc. Above the dotted line in fig. 8(b), the α phase is observed in the simulated fuels before dissolution with HNO_3 .

Comparing the results by chemical analysis with those by XRD there are good correlations for (1) the areas where the ϵ -phase and Mo, Ru and Pd are found, (2) the areas where the ZM-phase and Zr are found, and (3) the areas where the α -phase and Rh are found.

A similar comparison is also made at 2073 K. At high temperatures the ZM-phase shifts to higher oxygen potentials. This result is in good agreement with the fact that Zr is determined mainly in higher oxygen potentials (see fig. 7(b)). An other important tendency is that Pd and Sn are rarely found at all areas (see figs 6(b) and 7(b)). It should be noted that a few percent of U as well as Nd and Ba are determined in the residues when the treating temperature is higher than 2073 K, as seen in figs. 6(b) and 7(b). This means that not only U but also Pu in a spent LWR fuel has the possibility to be undissolved if a LWR fuel reaches to a higher temperature than 2073 K during irradiation.

Neither the perovskite phase ((Ba, Sr) ZrO_3) nor the scheelite phase ((Ba, Sr) MoO_4) are identified in residues by XRD. By chemical analysis Ba and Sr are seldom detected in residues (see figs. 4, 6, 7). These phases are thought to dissolve with 3M HNO_3 at 368 K.

As is clearly shown in figs. 3 and 4(a), both the amount of residue and Pd mole fraction show a distinct decrease between 1873 and 2073 K for the simulated fuel corresponding to 5% FIMA at a low oxygen potential controlled by 4% H_2 + 96% He. Considering other results that Pd is rarely detected at 2073 K but well detected at 1673 K (5% FIMA, fig. 6), and that neither Pd nor Sn are detected at 2073 K (10% FIMA, fig. 7), it appears that the cubic α -phase formed at high temperatures is easily dissolved in HNO_3 [5]. It may be possible that the ϵ -phase converts to the σ -phase to some extent with the formation of the α -phase in the fuel at high temperatures, therefore causing the Mo/Ru ratio in the residue to tend to increase from 1/1 to 3/2.

At high oxygen potentials controlled by He, the treating temperature does not affect the amount of

residue for burnup of 5% FIMA (fig. 3, curve 2) but does affect the composition. The mole fraction of Ru decreases while that of Mo increases with increasing temperature (fig. 4(b)). This phenomenon may be related to the change of oxygen potential for the Mo– MoO_2 equilibrium due to temperature change. The mole fraction of metallic Mo should increase with higher temperatures for the following two reasons: (1) Oxygen potential for the Mo– MoO_2 equilibrium increases with increasing temperatures and (2) oxygen potential controlled by He decreases at higher temperatures.

The maximum amount of residue is produced with the increasing oxygen potential, especially for high burnups as seen in fig. 5. The maximum may be attributed to the formation of the oxide phase identified as $ZrMo_2O_7(OH)_2(H_2O)_2$ [8] by XRD. It is natural to consider that this zirconium-molybdate should form in the solution during dissolution, because this compound is a hydrated one and is not identified in the fuel before dissolution. Lloyd and Campbell [9,10] found that a crystalline zirconium–molybdenum compound ($Zr/Mo = 1/2$) formed from dissolved solution of irradiated LWR fuel. They tentatively identified this as $ZrMo_2O_8(H_2O)_2$ but could find no reference to this recognized X-ray pattern. Although the formation condition was somewhat different, the zirconium-molybdate they found in the irradiated fuel may be the same compound in this investigation using the simulated fuel.

If the amount of residue is corrected for the amount of this oxide, $ZrMo_2O_7(OH)_2(H_2O)_2$, the dotted lines shown in fig. 5 are obtained. It appears that the corrected amount of residue decreases with increasing oxygen potential. This decrease again may be related to the oxygen potential for the Mo/ MoO_2 equilibrium. The amount of metallic Mo in the alloy phase should decrease at higher oxygen potentials, because Mo tends to be oxidized to MoO_2 at higher oxygen potentials. This assumption may also well explain the experimental results in which the Mo/Ru ratio decreases with the increase in oxygen potentials, because the oxygen potential for Ru/ RuO_2 is much higher than that for Mo/ MoO_2 .

Changes in burnup show a great effect on the amount of residue but does not greatly affect the elemental composition of the insoluble residue. The amount of metallic residue corrected for the zirconium-molybdate oxide phase, shown in fig. 5 by dotted lines, is less than 0.5 wt% for burnups below 10% FIMA, but increases markedly for burnups higher than 20% FIMA. This increase in the amount of insoluble residue for high burnups may be related to the fact that nuclear fission of ^{239}Pu plays a greater role on the amount of fission

products. From ²³⁹Pu fissions, a larger amount of Ru and Rh, which tend to form an alloy phase, is formed in the spent fuel.

5. Conclusion

- (1) The amount of residue decreased with an increase in temperatures of heat treatment above 1873 K. This is attributed to the fact that the α -phase formed at high temperatures is apt to be easily dissolved in HNO₃.
- (2) The amount of metallic residue and the Mo/Ru ratio decreased with the increase in oxygen potentials. The oxygen potential for the Mo–MoO₂ equilibrium may be related to these results.
- (3) The amount of metallic residue increased markedly for burnups greater than 20% FIMA. The increased amounts of Ru and Rh in the simulated fuel may be responsible for this result.
- (4) The formation of a zirconium-molybdate hydrated compound, ZrMo₂O₇(OH)₂(H₂O)₂, was identified only in the residue. This phase is possibly formed during the dissolution process.

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